

SchoolKing Protocol

MVP Web3 Descentralizado

Pedro Victor Sousa e Silva
WEB 3.0

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Breve Resumo do Projeto

O projeto consiste em um MVP Web3 desenvolvido em Solidity na rede Sepolia, integrando token ERC20, NFT ERC721, staking, DAO e oracle Chainlink.

Arquiterura do Sistema MVP

Componente	Função
ERC20	moeda do protocolo
NFT	certificado/conquista
Staking	travamento de tokens
DAO	votação
Oracle	preço ETH/USD

Contratos do Projeto

SchoolToken

0x66B6AEed8994cb2edF7663FAF64ABF534D6659a53

[SchoolToken Etherscan](#)

SchoolNFT

0x062f3ae79076c5a40a22e6C39aF96f2a7eb96cC0

[SchoolNFT Etherscan](#)

Staking

0x0f6CFE354423515533913C0F8d567DA0AabcEA8f

[Staking Etherscan](#)

DAO

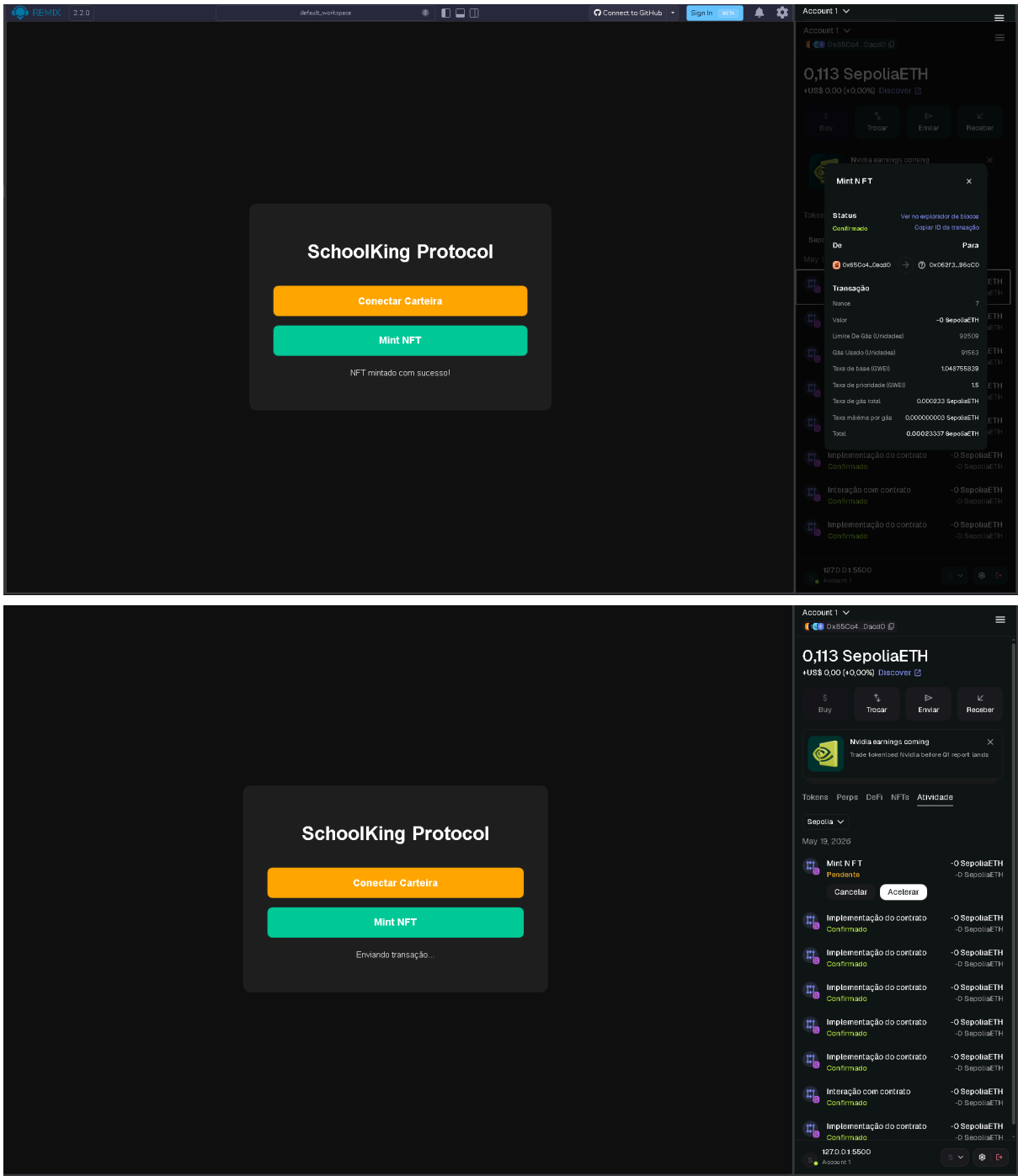
0xBaB0F05b90fe40F3C15Efc19bBaB286128D16a4f

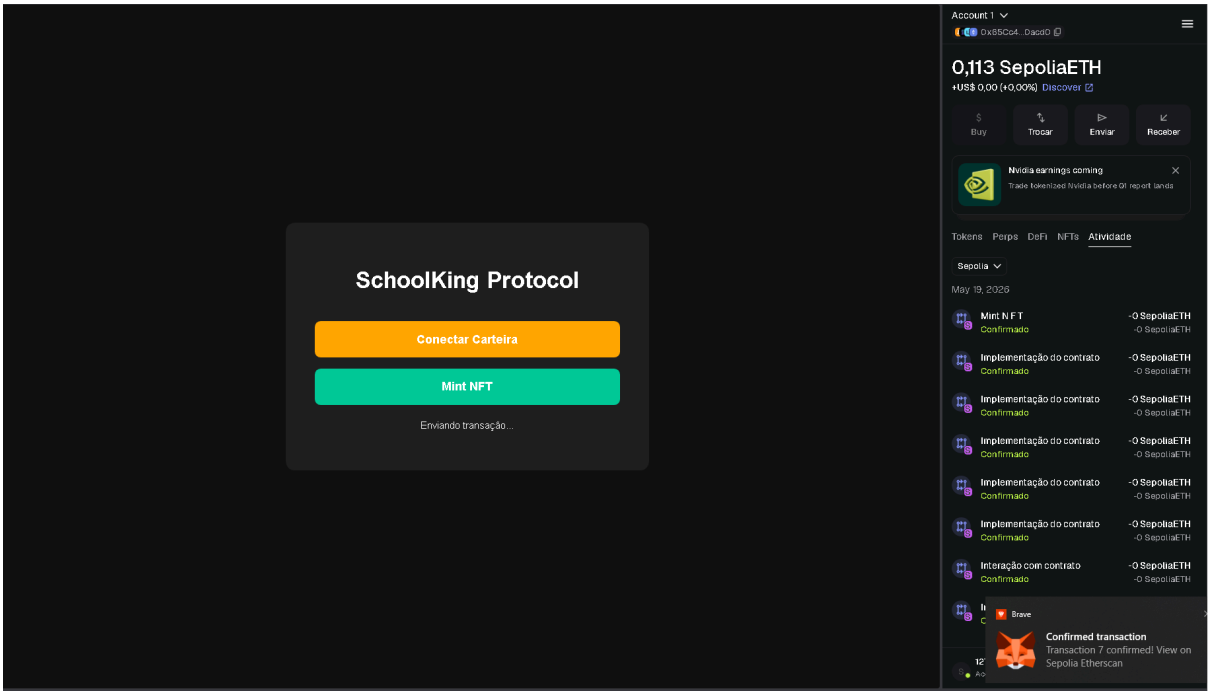
Oracle

0x08Fc5E82617ba54bcA55627AA7E8881a2E3F9fe5

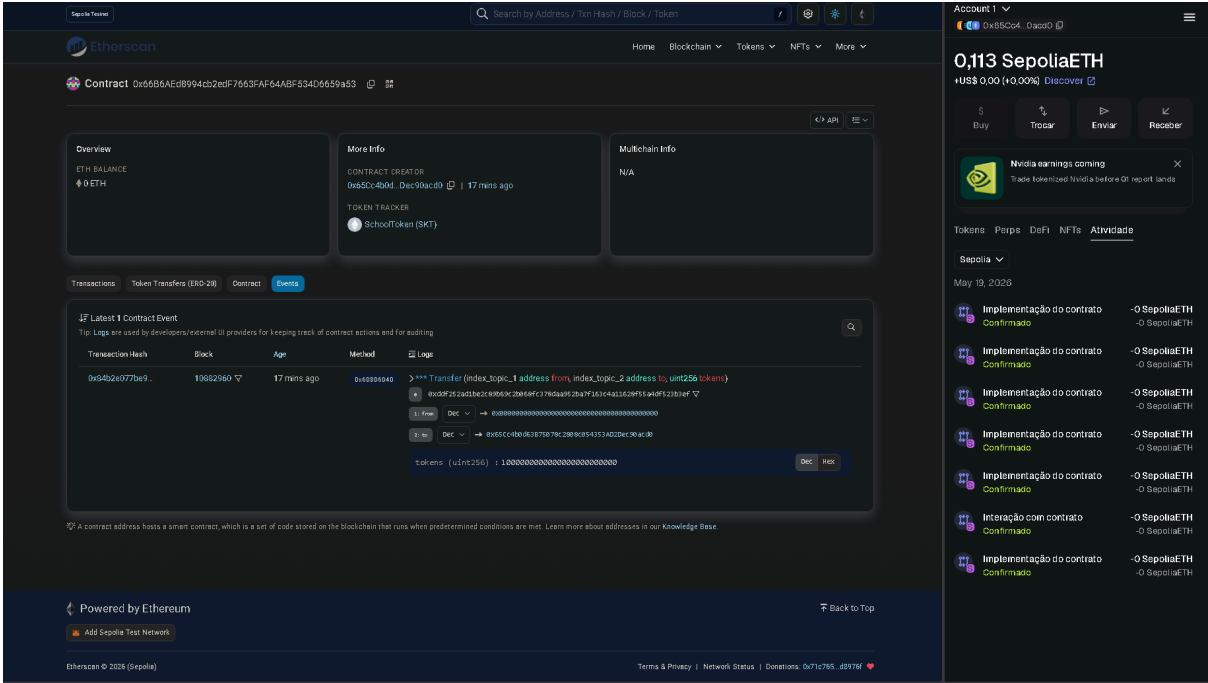
Prints do Projeto

Deploy completo do MVP (Front-end + Metamask + NFT Mint)





Deploy real em testnet



Integração NFT + blockchain

The screenshot displays the Remix IDE interface with the following components:

- Left Panel (Environment):** Lists deployed contracts including SchoolNFT, Staking, DAO, and Oracle, each with its address and a 'Details' link.
- Center Panel (Console):** Shows the Solidity code for the SchoolNFT contract. The code includes a license identifier, pragma statement, import, and contract definition with a constructor and a mintNFT function.
- Right Panel (Account):** Displays the account balance of 0.113 SepoliaETH and a list of recent transactions, all showing 'Implementação do contrato' (Contract implementation) for the SchoolNFT contract.
- Bottom Panel (Explain contract):** Provides a detailed explanation of the contract's functionality, including verification status and a list of transactions.

Staking funcional

The screenshot displays the Remix IDE interface with the following components:

- Left Panel (Environment):** Lists deployed contracts including Staking, SchoolNFT, DAO, and Oracle, each with its address and a 'Details' link.
- Center Panel (Console):** Shows the Solidity code for the Staking contract. The code includes a license identifier, pragma statement, import, and contract definition with a constructor and a mintNFT function.
- Right Panel (Account):** Displays the account balance of 0.113 SepoliaETH and a list of recent transactions, all showing 'Implementação do contrato' (Contract implementation) for the Staking contract.
- Bottom Panel (Explain contract):** Provides a detailed explanation of the contract's functionality, including verification status and a list of transactions.

DAO funcionando

The screenshot displays the Remix IDE interface with the following components:

- Left Panel (Environment):** Lists deployed contracts including `Account18`, `SchoolNFT`, `Staking`, `DAO`, and `Oracle`. The `DAO` contract is highlighted.
- Center Panel (Compiler):** Shows the Solidity code for `SchoolNFT.sol`. The code defines a `SchoolNFT` contract inheriting from `ERC721`, with a `mintNFT` function. The status bar indicates "Contract deployed. Checking explorers for registration..."
- Right Panel (Transactions):** Displays a list of transactions. The `DAO` transaction is highlighted, showing its details.
- Bottom Panel (Explain contract):** Provides a detailed explanation of the `DAO` contract, including its functions and the verification status.

The `DAO` transaction details are as follows:

Transaction	From	To	Value	Gas Used	Gas Price	Gas Cost	Transaction Fee	Transaction Hash
DAO	0x18...164F	0x18...164F	0.113 SepoliaETH	546422	540861	0.093930072	15 ETH	0x0014912 SepoliaETH

Integração Chainlink/oracle

The screenshot displays the Remix IDE interface with the following components:

- Left Panel (Environment):** Lists deployed contracts including `Account18`, `SchoolNFT`, `Staking`, `DAO`, and `Oracle`. The `DAO` contract is highlighted.
- Center Panel (Compiler):** Shows the Solidity code for `SchoolNFT.sol`. The code defines a `SchoolNFT` contract inheriting from `ERC721`, with a `mintNFT` function. The status bar indicates "Contract deployed. Checking explorers for registration..."
- Right Panel (Transactions):** Displays a list of transactions. The `DAO` transaction is highlighted, showing its details.
- Bottom Panel (Explain contract):** Provides a detailed explanation of the `DAO` contract, including its functions and the verification status.

The `DAO` transaction details are as follows:

Transaction	From	To	Value	Gas Used	Gas Price	Gas Cost	Transaction Fee	Transaction Hash
DAO	0x18...164F	0x18...164F	0.113 SepoliaETH	212399	10321624	0.000538 SepoliaETH	15 ETH	0.00053763 SepoliaETH

Transação blockchain real

The screenshot shows the Remix IDE interface. The central editor displays a Solidity contract with the following code:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 interface AggregatorV3Interface {
5
6     function latestRoundData()
7         external
8         view
9         returns (
10             uint80,
11             int256,
12             uint256,
13             uint256,
14             uint80
15         );
16 }
17
18 contract Oracle {
19     AggregatorV3Interface internal priceFeed;
20
21     constructor(address feedAddress) {
22         priceFeed =
23             AggregatorV3Interface(feedAddress);
24     }
25
26     function getLatestPrice()
27         public
28         view
29         returns (int256)
30     {
31         (
32             ,
33             ,
34             ,
35             ,
36             )
37     }
38 }
```

The left sidebar shows the 'Environment' with various accounts and transactions. The right sidebar shows the '0.113 SepoliaETH' balance and a list of transactions. The bottom status bar indicates the contract is deployed.

Integração com Front-end

Segurança do Projeto

Foram utilizadas bibliotecas OpenZeppelin e proteção ReentrancyGuard utilizando Solidity ^0.8.x.

Conclusão

O MVP demonstrou total integração funcional entre frontend Web3, contratos inteligentes e blockchain Ethereum Sepolia.